

STRATA: A Stratified Process-Mining Framework for HVAC Fault Detection and Diagnosis

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Thompson Rivers University — UREAP 2026

September 2026

Abstract

We report STRATA, a stratified process-mining framework for fault detection and diagnosis in building HVAC systems, and the falsification protocol under which it was built. One fault-free year of ordinary trend data and one configuration file naming a building’s sensors and control rules are abstracted, along ASHRAE Guideline 36 sequences, into an event log; process models are discovered at device and unit strata; and eight calibrated, physically interpretable channels, every threshold from the fault-free year and every reported rate out-of-sample, score each day against that channel’s own measured noise floor. On five public LBNL benchmark systems, two never previously benchmarked, the detector catches 158 of 175 seeded fault scenarios (14 of 14, 26 of 30, 25 of 29, 50 of 55 and 43 of 47; the single-duct count is 13 after a branch adjudication) at 26 false-alarm days in 480 held-out fault-free days, at a median of one day. It names the correct terminal unit in 37 of the 51 detections in the attribution universe and a wrong one in none, the right subsystem in 85 % of detections and the exact component in 71 %. A principal-component baseline detects 61 of the same 73 scenarios to STRATA’s 64, inside overlapping intervals, and cannot name the device. The false-alarm budget holds under injected field conditions on all five systems and on one measured rooftop unit (6.1 % of held-out days). The process-mining structure carries localisation and detections — the device stratum names the unit, and the absence, frequency and oscillation channels over the event log carry detections no rule names — while conformance checking of the discovered nets alone detects no fault the other channels miss: a pre-registered result, reported as a finding. Held to STRATA’s own gate, a third-party implementation of the Guideline 36 rule battery is within one detection and one false-alarm day on the single-duct air handler, and detects 8 of 30 and 9 of 29 on the terminal units against STRATA’s 26 and 25, at no false-alarm day against eight and nine. Thirty pre-registered experiments, ten adverse outcomes honoured and eight machine-verified benchmark errata are reported; every number regenerates from the public repository.

1 Introduction

Buildings account for approximately 34 % of global energy-related carbon dioxide emissions, HVAC carries the dominant share, and 15–30 % of HVAC energy is estimated to be wasted by undetected faults [5, 39]. The first large-scale characterisation, over 60,000 pieces of equipment in 317 commercial buildings, found 40 % of air handling units faulted on any given day [10] — buildings that run automation systems with alarms and rule libraries, so the gap is in the kind of fault such tools can see. Seven of the ten most common air-handler faults are behavioural — wrong schedules, unreleased overrides, missed resets, broken economiser sequences — deviations in the ordering of operating states, which a threshold or a pre-programmed rule sees only if someone wrote that rule

in advance [21]. The dominant research methodology does not model the sequence either: across 211 AI-based HVAC fault-detection studies, benchmark accuracies saturate at 97–99 % [5, 38, 43], the interpretable line explains at the level of input features [12, 51] rather than of what the building should have been doing, and an audit of 65 primary studies found two that shared code links, one broken [31].

STRATA changes the model class. Process mining discovers a Petri-net model of expected behaviour from a discrete event log and checks observed behaviour against it [44]; we derive the log from sensor data along Guideline 36 control sequences and discover models at device and unit strata. The framework as evaluated is a detector of eight calibrated, physically interpretable channels over that stratified structure, each reporting against its own measured noise floor, and its output is a work order: which unit, which relationship, by how much, and how often that channel has cried wolf on healthy days. The place of process mining in it is measured rather than assumed: the strata, the event abstraction, and the absence, frequency and oscillation channels that read the log carry the localisation and a share of the detections; conformance checking of the discovered nets alone, the founding hypothesis, detects no fault the other channels miss (Section 6.1). Process mining helps to get detection; it is not detecting on its own.

Four contributions follow: the framework and its measured performance on five public benchmark systems, reported at a stated false-alarm budget with the implicated device named; the evaluation protocol — pre-registration with binding falsifiers, adversarial recomputation before any number is quoted, and tests that pin published values, fired falsifiers and disclosure obligations; the pre-registered finding about what process mining contributes on this domain, its first application to building HVAC fault detection to our knowledge, corroborated by the one systematic review whose scope could have captured such work [6]; and eight machine-verified errata in the benchmark datasets.

2 Research Aim and Objectives

The question this research set out to answer is whether process models discovered from fault-free operation can detect faults in building HVAC systems interpretably and portably, and what the framework built around them achieves. STRATA is meant to consume a building’s ordinary one-minute trend data and one configuration file, learn from a fault-free period what healthy operation looks like, and every day name the days that break a physical relationship the healthy period established; onboarding a new equipment type should be a configuration exercise, not a software project. Six objectives follow, each with its status.

1. Abstract events from sensor data with control-sequence rules grounded in Guideline 36, holding fault-signature events out of model discovery. Done on seven LBNL datasets by configuration alone; the circularity the partition prevents was measured (Section 4). Met.
2. Discover models and locate invariants at device, unit and cross-system strata, and establish what they contribute. Done on every system, the contribution measured under pre-registration on 199 scenarios across six systems (Section 6.1). Met, in the form the measurement required: the strata and the log carry localisation and detections; the nets’ order structure does not.
3. Detect against a measured noise floor: eight channels, every threshold from the fault-free year, a three-way calendar split so every false-alarm rate is out-of-sample. Met; the budget held under injected field conditions and on one measured building.
4. Benchmark against known answers: 158 of 175 seeded scenarios at 26 false-alarm days in 480; 64 of 73 against an audited principal-component baseline’s 61; and a pre-registered comparison

against a third-party Guideline 36 battery, which fired its own falsifier (Section 6.2). All three parts executed; the improvement claim over current practice is withdrawn on the air handler and stands on the terminal units as a coverage difference bought with false alarms.

5. Make every claim checkable: thirty pre-registered experiments (Appendix A), ten adverse results honoured, every headline number regenerated under a regression gate, a test suite that fails the build if a published caveat is dropped. Met.
6. Release the work openly: repository, configurations, artefacts and reproduction path are public, with eight machine-verified dataset defects (Section 8). Met.

Two goals are future work: a hybrid extension feeding process-mining features to a classifier, deferred until the conformance finding was settled, and deployment on an operating building, for which the coverage audit of Section 6 is the checklist.

3 Related Work

Machine-learning HVAC fault detection is well surveyed [5]: hybrid pipelines reach 98 % on RP-1312 data [43], deep hybrids 97 % on real buildings [38], and the interpretable line explains at the level of input features [12, 51]. Unlabelled buildings are handled by self-supervised and autoencoder methods whose deviation signal is a scalar score [1, 13], and cross-building generalisation by adapting learned weights [15, 56]; what STRATA transfers is a configuration file, a different operation. Process mining [44] used conformance checking as a trace-level anomaly detector [4, 40] before neural sequence models overtook it [33], and has succeeded in adjacent sensor-driven domains — robotic-arm diagnosis [49], alignment features fed to classifiers [50], industrial IoT [23], and pasteurisation checked against HACCP control points [29], the closest analogue to Guideline 36 sequences — with event abstraction as the common barrier [24, 30, 46]. Building HVAC is absent from every review whose scope could have caught it [2, 5, 6, 20]; construction and facility-management applications concern human workflows [28], and the hand-designed Petri nets that have modelled HVAC for two decades [11, 17, 48, 53] are authored, not discovered from a log. A precedent for the shape of our conformance finding exists in intrusion detection [55]. Fault detection in real buildings is rule-based: commercial platforms ship libraries of named faults [20], Guideline 36’s conditions serve as reference for learned diagnostics [37], and the LBNL fault-correction line field-tested seven hand-coded algorithms over 225 equipment items [39]. Every such rule detects only what was enumerated in advance; STRATA is complementary, learning from the building’s own fault-free operation where behaviour is invariant and carrying a measured false-alarm budget per channel, on the trend data the commercial system already records [19].

4 Method

STRATA (Figure 1; code and artefacts at <https://github.com/iyassh/strata>) consumes one fault-free year of trend data and a single configuration file naming the building’s sensors and control rules — the only site-specific artefact; no code change was required to onboard any system reported here.

Event abstraction and the alphabet wall. Continuous streams are converted into a discrete event log by control-sequence rules grounded in Guideline 36 [24, 46], then partitioned into two alphabets that are never mixed: state events, neutral facts about operation and the only events admitted to discovery, and fault-signature events, which encode engineering suspicion and are withheld. The partition exists because of circularity, the process-mining form of data leakage [41, 52]: a model

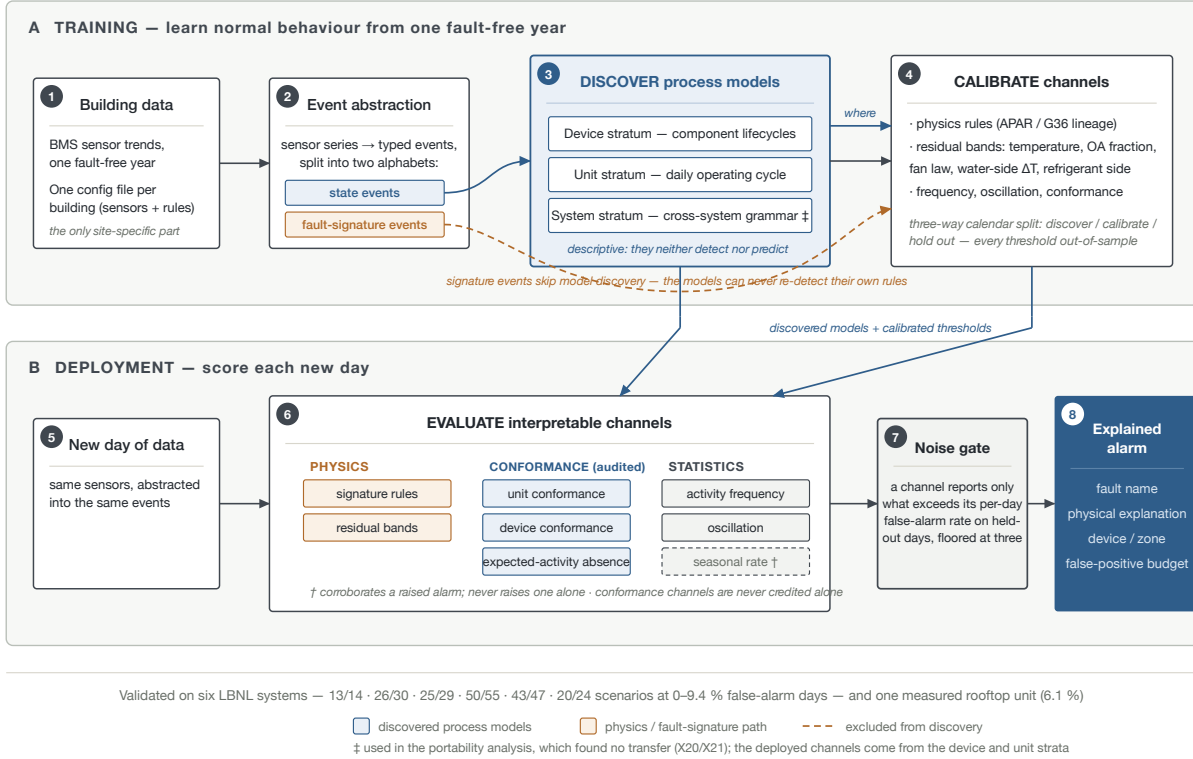


Figure 1: The framework as evaluated. Training (top): one fault-free year and one configuration file; events are split into state and fault-signature alphabets; process models are discovered from state events only; every channel’s threshold comes from a three-way calendar split (discover, calibrate, hold out), so every reported false-alarm rate is out-of-sample. Deployment (bottom): each day is scored by physics, statistical and conformance channels; a channel reports only what exceeds its own per-day false-alarm rate on the held-out days it could evaluate, floored at three days; what survives reaches the operator with a name, a physical explanation, the device and the stated budget.

discovered over an alphabet that names faults will later “detect” them by recognising the project’s own rules. Against the committed artefact of the pre-partition prototype on the single-duct system, the model channel’s apparent recall rises from 1.8 % to 5.2 % (81 against 238 days) without the partition while every combined-detector number is identical (3,152 true positives, 5 false positives): a measured 3.5-point bound on model self-confirmation, and a tested invariant now enforces the wall.

Discovery at three strata. Petri nets are discovered from the state log with the inductive miner [26, 27] in pm4py [3] at three granularities: the device stratum models an actuator’s lifecycle, the unit stratum the air handler’s daily cycle, and a cross-system grammar supports the transfer analysis. Whether a net encodes order at all is what precision metrics address [7, 32, 42, 47]; we use a direct reversal probe, every trace reversed and re-scored. The single-duct net falls from 0.8861 to 0.5526 under reversal; the series fan-powered net scores identically to four decimals (0.9533) and the parallel net shifts by 0.0005 (0.9021 against 0.9026), so neither carries ordering information in its fitness.

Table 1: The eight channels as deployed. Every threshold is set from the fault-free year alone; “alphabet” names which side of the wall the channel reads. The rate channel is advisory: reported, never counted in the deployed union.

Channel	Alphabet	A day is flagged when	Threshold source	In union
Rules	signature	a Guideline 36 / APAR-lineage rule fires during operation	fixed bands from the sequence documentation	yes
Residual	raw signals	a per-mode physical residual leaves its healthy band	[min, max] of training days, widened to the 0.5 °F sensor floor	yes
Model conf.	state	the day’s trace aligns to the unit net below threshold	1 % quantile of per-day fitness, calibration slice	yes
Device conf.	state	an actuator’s lifecycle trace aligns to its net below threshold	as above, per device net	yes
Absence	state	a device active on scheduled training days produces no event on a scheduled day	each device’s healthy silence rate	yes
Frequency	state	a daily activity count leaves its healthy range	[min, max] over training days	yes
Oscillation	raw signals	an actuator’s daily direction reversals leave their range	[min, max] over training days	yes
Rate	state	a 30-day active-day rate leaves the month-matched healthy range	training-year windows	advisory

Calibrated channels and the noise gate. Eight channels (Table 1) evaluate each day independently. Rules are physics rules of the NIST APAR lineage naming direct contradictions; residual bands encode relationships that must hold under known modes (with the cooling coil off, supply minus mixed temperature equals the fan heat rise); model and device conformance are alignment cost [8,45] against the unit and device nets; absence flags a scheduled activity that did not occur; frequency and oscillation are bands on activity counts and reversals; a seasonal rate channel corroborates but may not alarm alone. Calibration uses the fault-free year only, at a false-positive quantile of 0.01 against a calendar-stratified holdout of the last eight days of each month; no threshold sees fault data, and every rule needs a positive control, because silence is not competence (a position-based leak rule was structurally deaf on the fan-powered units and was replaced by a waterside flow rule). A channel then reports only what exceeds its own noise floor, tested as a binomial at $p < 10^{-3}$ against that channel’s false-alarm rate on the held-out days it could evaluate, floored at three, and thresholds are floored at sensor physics: an early result was retracted to at most two detection-days when a 0.5 °F precision floor showed the “detections” had cleared a 0.22 °F band by 0.02–0.03 °F. What survives reaches the operator with the implicated device, the evidence and the channel’s budget (Figure 2).

5 Verification as a First-Class Result

Every experiment is pre-registered: the hypothesis and the observation that would refute it are committed before the experiment runs, in the spirit of the hypothesis-testing discipline urged for process mining [36] and of pre-registration templates for predictive modelling [22, 35]. A falsifier that fires is published as a finding: thirty experiments were run this way (Appendix A), ten went against the project and all ten are reported (Section 7). The discipline exists because of a failure mode we observed in ourselves — one phase report drafted the day of its discovery contained six overclaims, all caught by audit within hours — and it catches drift: a robustness experiment scored through the importable library detected five dual-duct scenarios the scorecard did not, because the

library lacked a noise-floor repair the scripts had received; a guard now pins the two.

Before any result is computed, a gate battery verifies file integrity, timestamp monotonicity, calendar consistency and rotation, and hashes every derived log so duplicate scenarios cannot pose as independent evidence; a healthy-silence gate confirms each rule set is silent on the fault-free year before it may score anything, and three of the errata of Section 8 were found by this battery. Guards run at three levels — more than a hundred tests on every commit, a regeneration gate that re-derives every committed artefact and diffs it (the single-duct benchmark must reproduce byte-identically), and every validated dataset frozen as a fixture — and the suite pins three things not ordinarily thought testable: published values, fired falsifiers (the sampling-rate finding is its four refutations), and disclosure obligations, so that a regeneration which drops a published caveat fails the build exactly as a lost feature would. Continuous integration has guarded that a result regenerates [25]; we have not found it reported guarding that a caveat survives.

Before any number is quoted, it is independently recomputed from raw data by an agent briefed to find it wrong — AI coding agents in this project, denied the answer key where a test was sealed. The method’s value is what it caught: a baseline detecting file provenance rather than faults; one of fourteen single-duct detections that was the dataset’s configuration branch, not the fault (14/14 naive, 13/14 adjudicated); a day-level significance test resting on an impossible null, retracted and replaced by the binomial against each channel’s own noise; detections below sensor precision; and a completed comparison against commercial practice, voided twice before the execution quoted in Section 6.2. None would have been caught by a test suite, because in each case the code did what it was written to do. Thirty-two numbered lessons record each failure with the rule it produced; the standing one is that a lesson is not closed until a test mechanically enforces it. Simulation flatters, and one measured stream now bounds that: on an LBNL field rooftop unit (182 fault-free days over two summers, 25 points, no schedule point), onboarded by configuration plus one derived fan-on column and two logged silence iterations, the deployed detector’s false-alarm rate is 3 of 49 held-out days (6.1 %) with every channel inside budget; its one fault case, a 40 % refrigerant undercharge, is not seen and is reported as a case, not a rate.

6 Results

Evaluation began with three public LBNL datasets [18,19]: a single-duct air handler (SDAHU) and parallel and series fan-powered units (PFPU, SFPU), the latter two never previously benchmarked; a dual-duct air handler (DDAHU) and a fan coil unit (FCU) were onboarded blind afterwards, and a simulated and a measured rooftop unit complete Table 2. Scoring follows day-level, scenario-as-unit conventions [14,54]; each scenario is a year-long single-fault file, and every false-alarm count is on the 96 held-out fault-free days (the last eight of each month), out-of-sample since the three-way split.

Detection at a stated budget. Under the final gate the detector catches 158 of 175 seeded scenarios on the five simulated systems at 26 false-alarm days in 480, the deployed union at 1.0, 8.3, 9.4, 3.1 and 5.2 % of held-out days; coil fouling is the dominant miss on the four air-side datasets and is caught in full on the simulated rooftop unit, which records the refrigerant side. The median time to detection is one day on every system and the tail is short: on the single-duct unit 9 of the 13 adjudicated detections come on day 1 and 4 on day 2; on the fan-powered units all 26 and 25 come on day 1. The cry-wolf ratio, wrong alarms as a share of all alarms, is at most 0.13 % (0.03, 0.11, 0.12), small partly because observed fault-scenario days outnumber fault-free days 47–99 to 1. Pooled recall misleads for gated detectors: the coil-bias residual is physics-bound only inside

System (LBNL)	points	rules (state / sig.)	scored	detected	false alarms	healthy days
SDAHU, single-duct AHU	26	3 / 12	14	14 (13 adj.)	1 / 96	365
PFPU, parallel fan-powered box	109	13 / 45	30	26	8 / 96	365
SFPU, series fan-powered box	109	13 / 49	29	25	9 / 96	365
DDAHU, dual-duct AHU	114	15 / 56	55	50	3 / 96	365
FCU, fan coil unit	29	7 / 19	47	43	5 / 96	365
RTU, simulated rooftop unit	23 [†]	3 / 5	24	20	0 / 28	100
RTU, measured rooftop unit (Site 2)	26 [†]	2 / 7	1 case	—	3 / 49	182

Table 2: The seven LBNL datasets under the final gate: points mapped after the coverage audit, scenarios scored and detected, and out-of-sample false-alarm days on the held-out fault-free days. The measured unit carries one fault case and is used for its false-alarm rate only. [†]Includes one derived operating-state column.

System	columns mapped	detected, before → after	false alarms, before → after	lost
SDAHU	12 → 26 of 30	14 → 14 / 14	1 → 1	0
PFPU	56 → 109 of 109	22 → 26 / 30	7 → 8	0
SFPU	56 → 109 of 109	21 → 25 / 29	6 → 9	0
DDAHU	79 → 114 of 114	45 → 50 / 55	3 → 3	0
FCU	29 → 29 of 29	40 → 43 / 47	4 → 5	0
all	232 → 387 of 391	142 → 158 / 175	21 → 26 of 480	0

Table 3: The coverage audit: sixteen scenarios recovered, none lost, five false-alarm days added over 480 held-out days, the model and device rows identical on every system. Totals use the naive single-duct count.

a window (coil off, occupied, sustained 120 minutes) that exists on 137–164 days a year in this climate, and within it recall is 164/164, 164/164, 139/139 and 137/137 on the four bias scenarios; the apparent 39–47% ceiling is climate.

The coverage audit. The misses remaining after the statistical repair had been read as limits of what the datasets record; a read-only scan found that for six of the fourteen reachable misses the decisive signal was a recorded column no configuration mapped. A pre-registered series (X33a–e) audited every column of every system, mapped every usable one with a physically named reader (a residual, a rule, a gate or an oscillation signal), excluded the rest with a written reason a build-time guard enforces, took every band from the fault-free file, and left the state alphabet untouched, so the process-mining rows had to regenerate byte-identically (Table 3). Sixteen scenarios were recovered, none lost, five false-alarm days added over 480: airside reheat-coil fouling through the series fan’s power per cfm delivered (“same power, less air”), room-temperature biases through the return air falling away from the biased zone, the dual-duct hot-deck biases through the unbiased copies the mixing boxes record, and the fan coil’s damper leaks through the outdoor-air fraction. The seventeen remaining misses are coil fouling on every system, whose recorded values sit inside every healthy band on every column; one is visible but uncertifiable from a single healthy year. Two disclosures bound the claim: which columns to read was decided after the scan of the fault files, so the readers were fault-informed though every threshold was blind; and five of the sixteen recoveries sit less than one band width outside the healthy extremes. The series shows these misses were reachable from the recorded columns with blind thresholds, not that a blind onboarding would have reached them; the blind onboardings below are the honest prior for a new building.

Against a learned baseline. A principal-component baseline on squared prediction error, audited (it had been detecting file provenance through the columns of erratum E3) and scored on identical scenario universes, detects 61 of the 73 scenarios on the first three systems to STRATA’s

STRATA alarm -- series fan-powered unit, 2018-01-01 (day 1 of the fault)
 what zone primary-air flow not tracking its setpoint [channel: rules]
 where terminal unit TU_S, zone S
 evidence measured 715 CFM against a 200 CFM setpoint -- a 515 CFM excess
 sustained for 720 min while occupied (rule fires at 50 CFM held
 60 min); damper position 0.80, fault-free same-day value 0.32
 budget rules channel: 0 false-alarm days on the 96-day fault-free
 holdout; threshold set on training days only, holdout out-of-sample
 corrob. seasonal rate channel agrees (advisory; never alarms alone)
 seeded zone-S damper stuck at 80% -- location correct, symptom correct,
 detected on day 1 of 365

Figure 2: A worked alarm, every value read from the committed artefacts and the raw data. Scenario SFPV_VAVDMPRStuck_80%; the alarm names the symptom and its location, not the root cause.

64 (13/26/25 against 13/23/25), inside overlapping Wilson intervals ([78.2, 93.4] % against [73.4, 90.3] %); the three discordant scenarios are all STRATA's (an unstable damper and the parallel unit's moderate and severe airside fouling), none the baseline's, and an exact McNemar test on three pairs ($p = 0.25$) is an absence of power, not evidence of superiority. Nine scenarios are missed by both, eight of them reheat-coil fouling. On false alarms the baseline is lower on the terminal units (1/96 against 8/96 and 9/96) and within a day on the air handler (1/79 against 1/96), intervals overlapping throughout; it consumes 108 and 424 raw daily features where STRATA reads physically named relations between mapped columns. What the baseline cannot do is name the implicated device or state the evidence behind a flag.

Localisation and diagnosis. Every seeded zone fault in the benchmark occurs in the same zone, so attribution is reported as specificity, not discrimination. On the 51 detections on the two fan-powered units in the attribution universe (65 naive detections less the 14 single-duct ones, which have no device stratum), the alarm names the correct terminal unit in 37, names none in 13 and names a wrong unit in none, with one indeterminate; three wrong units were available in every case, so the zero is falsifiable. A pre-registered component test (X34) over all 158 detections found the carrying rule naming the seeded component exactly in 109 (69 %), the right subsystem in 22 more (14 %), the wrong subsystem in 20 (12.7 %) and no component in 7 (frequency- or oscillation-only detections); the bar for a component-level claim (wrong subsystem ≤ 10 %) was missed, so the claim is subsystem-level. Fourteen of the twenty wrong cases are on the single-zone fan coil unit and name the component that responds to the fault — the coil for a biased room sensor, the fan for a fouled coil — the first place a technician would look. A pre-registered diagnosis layer (X37) of three fault-agnostic ingredients — the sign of every significant residual, a consensus rule over redundant copies of one quantity, and a leave-one-out nearest-pattern vote for the fault family — resolved the five dual-duct damper cases the carrying rule had scored wrong (one mixing-box copy moves alone) and called 14 of 16 deck sensor biases at the deck, lifting exact attribution to 112 of 158 (71 %) and exact-or-subsystem to 85 %, with the wrong-subsystem rate at 10.1 %. It missed its own bars — the family is named from the signed pattern alone in 85 of 114 resolved detections (74.6 %, bar 80) with 44 unresolved (27.8 %, bar 15), because a stuck damper and a biased airflow sensor, a restricted filter and a fouled coil, a leaking and a stuck valve are pairs the recorded quantities cannot separate by sign — so it is not adopted. What the framework can honestly say is what it measured: zone right and never wrong, subsystem right in 85 %, exact component in 71 %.

Onboarding blind. The claim that a building is a configuration directory was tested on two system types the project had never seen, under pre-registrations written after the documentation and before any data file was opened. The dual-duct unit (55 faults in 15 families, two new) took 79 mappings, 15 state and 26 signature rules and one healthy-silence iteration — ten healthy days on which the cold-deck fan saturates, removed by a physical gate rather than a looser threshold — and scored 45 of 55 at 3 false-alarm days in 96 with no code change (50 after the coverage audit). The fan coil unit (47 scored faults in 17 families, four new) took 29 mappings and 7 state and 16 signature rules; its first run scored 47 of 47 at 28 false-alarm days and fired two falsifiers, because two scoring scripts written for the fan-powered units treated an event-less setback day as a scheduled model violation, and a pre-registered two-script repair restored 41 of 47 at 4 false-alarm days with the four earlier scorecards regenerating byte-identically (43 after the coverage audit). The second system type had been onboarded in about 1.5 hours from commit timestamps, 45 mappings and 35 rules; the third was derived from it in minutes.

Field conditions and calibration. On the shipped configurations of all five systems after the coverage audit (X35), field sensor error on temperatures, flows and positions with 5% feedback quantisation, change-of-value logging with stale repeats and gaps, and schedule shifts with holidays, each and all together, applied to the fault-free year and every fault file and re-fitted, kept the budget in every cell (worst 7 of 96, the parallel unit under the schedule shift; no system more than two above its clean count), cost at most one detection per system and condition, gained nothing, and reproduced the scorecard on the clean arm; of the five edge recoveries one was lost under noise where the pre-registration predicted at least two. A pre-registered alternative calibration (X36), bands at the 2nd and 98th percentiles of the training days instead of the extremes, about ten per cent narrower, gained no scenario and lost none, kept all five edge recoveries, and added thirteen residual false-alarm days over 480, four each on the fan-powered units; its budget falsifier fired and the deployed [min, max] calibration is unchanged. Earlier tests bound the same from the other side: under i.i.d. jitter at sensor precision the counts are unchanged and under a mirrored holdout they move by at most one, while the membership of the marginal scenarios shifts; and a single score-carrying contaminated training day collapses the two negative coil-bias rungs from 164 of 164 detections to none, the mild contamination source visible on 100% of its days (5/5, 13/13, 27/27) and the dangerous one on none of its 27.

6.1 What process mining carries, and what it does not

The stratified structure is what makes the alarm of Figure 2 possible: the device stratum is why an alarm names a terminal unit, and the single-duct system, which has none, names no device by construction. The event log carries detections of its own: the frequency channel measures count deviations two-sidedly and was significant on 12 of the 61 scenarios the scorecard then detected on the first three systems; the oscillation channel stands alone on the unstable-damper scenario, 349 detections against at most two from any hand-written rule; seven detections across the five systems are carried by the frequency or oscillation channel alone, instability faults and one fan restriction that no rule names; and on the dual-duct unit a day-one alarm belongs to the absence channel. A matched-rule control isolates what discovery contributed at design time: for each channel a hand-written equivalent was authored, and two reproduce their channel exactly (141 against 141 detections on the series unit, 206 against 206 on the parallel unit) — the discovered models surfaced invariants an expert then wrote as rules, which is how several deployed rules came to exist.

Conformance checking of the discovered nets is a different matter, and the result is measured. Reading the significance-gated channel credited with each detection from the committed scorecards,

no scenario on any of the six systems is detected only by the two conformance channels: they are credited with none of the single-duct or parallel-unit detections and appear in 7 of 25 on the series unit, never alone. Removing them changes no scorecard and lowers the series unit’s false-alarm rate from 9/96 to 4/96 and the parallel unit’s from 8/96 to 6/96; over 199 scenarios the one-sided 95 % upper bound on the sole-detection rate is 1.5 %. The log explains why: under waterside reheat-coil fouling the state-event log is identical to healthy to the minute (162.8 against 162.4 events per day), eight of the twelve scenarios then missed have a log indistinguishable from healthy at a 5 % tolerance, so no method that reads that log can detect them, and under a 50 % reheat-valve leak the log changes in timing and counts with its order unchanged, which control-flow conformance cannot see by construction. Three pre-registered extensions followed. A time-infused state model (X12) is significant on 11 of 73 scenarios and on none the deployed channels miss, never earlier than day 14 against day 1, at five extra false-alarm days per fan-powered unit. Healthy-derived band states for every actuator and flow (X14) enrich the alphabet by 6, 40 and 36 events; the single-duct model channel then fires on all 14 scenarios instead of none and one miss, moderate airside fouling of a series-unit reheat coil, becomes significant through a dose-monotone zone-damper signature, at 7, 18 and 21 false-alarm days in 96, over budget, so that falsifier fired. The enriched frequency channel alone (X15) keeps that one gain inside the budget under two splits, a candidate addition rather than an adopted one, and does not replicate on the dual-duct unit (X18).

A pre-registered positive control settled whether the channel can see order at all: injected violations (reversal, adjacent swaps, deletions, a deleted start, a duplicated block) scored on each system’s fault-free holdout with the deployed net and gate. The channel is three instruments. On the fan coil and dual-duct units a reversed day is flagged on 55 of 72 and 59 of 83 holdout days (AUC 0.99): the channel sees order and those units’ faults do not change it. On the single-duct unit a reversed day is separable on average (AUC 0.91) but the healthy holdout contains days that fit worse than a reversed one, so no in-budget threshold sees order. On the fan-powered units fitness does not move under any perturbation (AUC 0.36–0.51): blind by net structure. A seven-instrument grid (X30) — the inductive miner at two noise settings, token replay, the heuristics miner, the log skeleton, Declare constraints and the temporal profile — thresholded and gated as every channel is, detects none of the 33 scenarios the detector then missed at realised false-alarm rates of 0 to 9.7 %: the null is a property of the state-event log, not of the miner or the measure. Sixteen of those 33 were later recovered by the coverage audit, through columns no process model was ever given.

6.2 Against current practice: the falsifier fired

The comparison practitioners care about is against the rule-based detection deployed in commercial automation systems [20]. A third-party open-source implementation of the Guideline 36 fault conditions, open-fdd 4.4.1 [34], was imported at its installed defaults, air handlers graded with the standard’s fifteen conditions and terminal units with the library’s terminal-unit rules per zone, on the identical examination; the pre-registered refutation condition was that the battery matching STRATA’s count at equal or lower false alarms on any system would abandon the improvement claim there. Two executions were voided first, the second after two independent audits found six adapter defects and, in the design itself, a single gate on the pooled union of all rules, a tautology once that union flags every healthy day; an amendment recorded all of it before the third execution, which is quoted (Table 4).

Held to STRATA’s own treatment of its signature rules — keep the rules silent on the fault-free year and hold them to the same 3/365 null — the battery detects 12 of 14 on the single-duct air handler at one false-alarm day against STRATA’s 13 at one (its two misses are the 10 and

	SDAHU (14)	PFPU (30)	SFPU (29)
Battery, pooled gate (as first registered)	1 @ 49	0 @ 96	0 @ 96
Battery, per rule against its own holdout rate	12	30 [†]	29 [†]
Battery, rules silent on the healthy year, held to 3/365	12 @ 1	8 @ 0	9 @ 0
STRATA, deployed	13 @ 1	26 @ 8	25 @ 9

Table 4: The repaired Guideline 36 battery against STRATA: scenarios detected @ false-alarm days in 96. [†]Not detection: three air-handler conditions flag the identical day sets on the fault-free file and every fault file and pass only because each rule’s holdout rate sits below its full-year rate; excluding the self-detecting rules post hoc gives 15 of 30 and 17 of 29. STRATA’s single-duct count is 13 by the branch adjudication of erratum E5, which the battery’s count does not apply.

25% stuck dampers, seen only by a condition that is not silent), and 8 of 30 and 9 of 29 on the terminal units at no false-alarm day against STRATA’s 26 and 25 at eight and nine, its detections being the unstable room loop and the larger stuck and leaking valves and dampers. The falsifier fired on every system — on the air handler because equal false alarms is a match, on the terminal units because none is fewer — so the pre-registered improvement claim cannot be made. What the artefact supports is narrower: on the air handler a correctly wired Guideline 36 battery, silenced on the fault-free year, is within one detection and one false-alarm day of STRATA; on the terminal units STRATA detects three times as many seeded faults and pays eight or nine false-alarm days in 96 for them, and the difference is coverage, the per-zone residual readers of the coverage audit, not the gate. Four caveats travel with the numbers: the battery’s rule selection is judged on the whole fault-free year including the holdout, so its one false-alarm day is a post-selection count where STRATA’s is out of sample; the outdoor-air-bias file is adjudicated away on STRATA’s side only (removing it from both gives 11 of 13 against 13 of 13); the parameters are open-fdd’s installed defaults, tighter on every condition that fired than the Guideline 36 values the library cites; and the terminal rules carry fixed comfort bands that do not read the building’s setpoint columns.

7 What the Protocol Refuted or Withdrew

Ten results went against the project; each is reported with what survived. (1) Discovered models do not out-detect hand-written rules. On the series unit, zone-damper and airflow-bias faults erase the daily heating rhythm and the healthy net flags the absence; a one-line matched rule — occupied workday and heating never active, zero false alarms on 365 healthy days — covered 192 days against the model channel’s 135 (178 in the current artefact). The falsifier fired; the rule is a permanent ablation arm. (2) Behavioural structure does not transfer zero-shot. Fan-powered logs outscore their shuffled controls on the single-duct net by 0.15 and 0.18 in fitness, real shared structure, but the detection hypothesis yielded zero to one gated detection per rotation; portability is claimed at configuration level only. (3) The models’ detections were count deviations in disguise, surfacing through alignment cost; once the frequency channel measured them directly the conformance channels had nothing uniquely theirs (Section 6.1). (4) 14 of 14 became 13 of 14 when an audit found the single-duct fault-free file simulated on a different configuration branch than every fault file (erratum E5); under a branch-corrected band one scenario’s only channel produced zero flags on 148 window days. (5) A day-level significance test was retracted: its discordant cell was zero by construction; thirteen parallel-unit rows lost their stars and the null is now always the channel’s own noise. (6) Healthy silence does not survive resampling: configurations tuned at one-minute sampling gain 28 to 64 spurious healthy signature-days at fifteen minutes and one residual’s coverage drops by more than a quarter; four falsifiers fired, and the silence gate is re-run at the deployment sampling rate. (7) A pre-registered transfer test never had power. Whether model

ID	System	Defect	Consequence
E1	SDAHU	oa_bias: one run shipped four times under different severity labels; the labelled sensor-side bias never appears in the logged column.	Four labels index one run; the mechanism is not asserted.
E2	SDAHU	coi_leakage: one run shipped four times under different severity labels.	Four labels index one run; the family is scored once.
E3	SDAHU	A constant of about 401.86 added to SA_SP in the fault-free file and subtracted from SA_SPSPT in every fault file, ranges disjoint.	Either column separates healthy from fault by provenance; both excluded from all baselines.
E4	SFPU	SensorBias_RMTEMP_-2C: calendar rotated relative to its siblings.	Excluded; the SFPU universe is 29.
E5	SDAHU	The healthy file was simulated on a different configuration branch than every fault file (occupied damper floor 0.0 against 0.1, different schedules).	One detection was provenance; 14/14 naive, 13/14 adjudicated.
E6	FCU	Two minor airside-fouling files (cooling, heating) are byte-identical.	One label is wrong; scored once, the FCU universe is 47.
E7	PFPU, SFPU	Documentation places every seeded fault in the west zone; in every file only the south zone's columns move (57 of 60; three indeterminate).	Localisation scored against the documented zone would mark every correct alarm wrong; ground truth taken from the data.
E8	DDAHU	Static-pressure bias files: the deck column is unchanged and the box statics shift by one tenth of the labelled bias.	Severity ladders built from these labels are off by ten; detections unaffected.

Table 5: Eight machine-verified defects in the LBNL benchmark datasets.

support predicts where a transplanted rule holds was tested with quantity, predictions and falsifiers sealed; but the model quantity is constant within a building (0.000, 0.455, 0.978), so nine cells were three buildings pseudo-replicated and the valid null has minimum $p = 1/3! = 0.167$ against a criterion of 0.05. Three dated amendments record it; the reported $p = 0.20$ is not a measurement and no inference is drawn. (8) A fault-informed coil-effectiveness residual for the fouling misses was significant on one of twelve fouling scenarios, already detected; its falsifier fired. The positive control is LBNL's simulated rooftop unit, the one dataset recording the refrigerant side, where the unchanged detector catches every condenser- and evaporator-fouling severity down to 10% (10 of 10, monotone) at 0 false-alarm days in 28. (9) Richer vocabulary buys detections at false alarms the budget forbids (X14, Section 6.1). (10) The powered transfer re-run passed for the wrong reason. With four scorable buildings the pre-registered identity test gives $\rho = -1$, exact $p = 1/24 = 0.042$, and the artefact records the pass as licensed; the paper declines it because a discovery-free control, the raw heating-day fraction, orders the buildings identically, and the non-circular variant, a net fitted on one building reading another's year (X21), orders them no better ($\rho = -0.6$).

8 Defects in the Benchmark Datasets

Auditing the evaluation produced eight defects in the LBNL benchmark datasets (Table 5), each with machine-verifiable evidence and a reproducing command in the repository's ERRATA.md; the creators' own account of the models' fidelity limits [9] is the closest prior, and these are byte- and label-level findings. The most consequential for other researchers is E3: a single threshold on either static-pressure column, chosen from the fault-free file alone, classifies all 365 fault-free days and all 7,150 fault days correctly, so any model given those columns identifies provenance perfectly without reference to any fault. Published results on this dataset that used the full column set should be read with that in mind.

9 Limitations

All detection results come from simulation, and simulation flatters; the field-condition injections and the one measured stream narrow the gap without closing it. The measured rooftop unit supports a false-alarm rate and one unseen fault case, nothing more, and until deployment on an operating building is complete the framework is validated in the laboratory. The contamination result bears on that gap directly, because a real building’s baseline year is not fault-free by construction.

Two reporting decisions are disclosed. The deployed rates treat the seasonal rate channel as corroborating evidence that may not alarm alone; the naive union of all eight channels false-alarms on 15.6 % of held-out days on the single-duct system, nearly all of it that channel, the demotion was adopted after those figures were known and costs no detection, and readers comparing against an undemoted union should use 15.6 %. Second, until the three-way split the conformance channels’ holdout rows were calibration targets rather than measurements; the split measured that debt (the fan-powered budgets moved from 5.2 and 4.2 to 7.3 and 6.2 %), and making the residual channel’s null a per-day rate then removed seven residual-only detections, a loss that fired a pre-registered falsifier and is reported as a detector change. Every reported row is now out-of-sample, though every threshold comes from the same fault-free year.

The noise-floor gate applies no correction across channels and scenarios: on the order of 1,600 binomial tests at $\alpha = 10^{-3}$ give a family-wise error near 0.8 under independence, about 1.6 false cells across the study, and per-day observations within a scenario are autocorrelated. All counts are single-run figures on one simulated year per system. Gradually developing faults are detected slowly, the expected cost of calibrating below a noise floor. The component-level attribution claim and the diagnosis layer both missed their pre-registered bars; the alarm names a symptom and a location, not a root cause. The hybrid extension is deferred deliberately: its output would be an anomaly score, and the test worth running is whether it detects more at an equal budget while still naming the device. Six simulated systems across five equipment classes demonstrate configuration-level portability, not generality across commercial HVAC; a 2026 collection of eight further labelled AHU datasets [16] is the obvious next target.

10 Conclusion

STRATA does what its intent asks on the benchmarks it was tested on: from one fault-free year and one configuration file it detects 158 of 175 seeded faults on five public systems at 26 false-alarm days in 480, most on the first day, names the terminal unit correctly and never wrongly, the subsystem in 85 % of detections and the exact component in 71 %, and issues each alarm with the evidence behind it and the channel’s own false-alarm budget — properties a scalar anomaly score of equal detecting power cannot supply. Its budget held under injected field conditions on every system and on one measured building. Process mining is what makes that structure possible, and its contribution is now measured rather than assumed: the strata name the device, the event abstraction from the control sequence is the vocabulary, and the absence, frequency and oscillation channels over the log carry detections no rule names; conformance checking of the discovered nets alone detects no fault the other channels miss, and the log diagnosis, the positive control and the seven-instrument grid say why. Process mining helps to get detection; it is not detecting on its own.

The protocol is offered as a contribution in its own right: thirty pre-registered experiments, ten adverse outcomes honoured, a completed comparison against commercial practice voided twice by audit before the execution that was quoted, and eight defects in a public benchmark documented

with reproducible evidence. In a subfield where independent verification is rare, a result that has survived a genuine attempt at refutation is worth more than a larger number that has not.

Acknowledgements

The author thanks Dr. Anthony Aighobahi and Dr. Mridula Sharma for supervision throughout this project. AI coding assistants were used for implementation, analysis scripting and adversarial auditing of results; all experimental designs, pre-registrations and claims were reviewed and approved by the author.

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A Every pre-registered experiment

Tables 6 and 7 list every pre-registered experiment in the order it was run, with its prediction, outcome and what fired. Each row resolves to a ledger under outputs/ or, where noted in the repository’s research log, to the phase report; amendments are folded into the row they amend.

ID	Experiment	Prediction	Outcome	Falsifier
—	Do discovered unit models out-detect a hand-written rule?	Model channel covers more fault days than any matched rule	Matched rule 192 days against the model channel's 135 (178 now) at 0 of 365 healthy false alarms	Fired
—	Does behavioural structure transfer zero-shot?	Source-fitted bands detect on the target	0 of 14 single-duct scenarios under the confidence-bound gate	Fired
—	Does model support predict where a transplanted rule holds? (three amendments)	$\rho < 0$, permutation $p < 0.05$ over nine cells	$\rho = -0.69$, $p = 0.200$; valid null over three buildings has minimum $p = 0.167$	F1, F2 fired, then voided; no power
X5	Are severity ladders monotone in dose?	No ladder significantly anti-monotone	Series-unit fouling $\rho = +0.87$; none anti-monotone	None
X7	Does healthy silence survive 15-minute resampling?	Signature days and residual coverage preserved	Healthy years gain 28, 64 and 64 signature days; one residual's coverage falls 31 to 1	F-X7.a (three systems), F-X7.b
X8	Does training contamination break extreme-value calibration?	Breakdown at small contamination; mild sources visible	One contaminated day collapses both negative coil-bias rungs, 164 of 164 to 0	None
X9/X10	Robust to sensor jitter and to the holdout split?	Detections change by at most 3; false alarms under 10 %	Unchanged at $1\times$ jitter, within one on the mirrored split; 0–4 false alarms of 96	None
X11	Is the single-duct OA-bias detection the fault or the branch?	Five estimators agree; the branch offset explains the flags	Offset $+1.06^\circ\text{F}$; 0 flags on 148 window days; 14/14 becomes 13/14	F-X11.d (run 1); leg re-specified
X12	Does the time perspective recover missed scenarios?	Duration channel newly detects missed scenarios	11 significant, 0 newly detected	None; P3 failed
X13	Does a coil-effectiveness residual separate reheat fouling?	At least 6 of 9 fouling scenarios significant	1 of 12, already detected elsewhere	F-X13.b
X14	Was the model starved of vocabulary?	One new detection inside the budget	One new; model channel fires on 14 scenarios; 7, 18, 21 false alarms of 96	F-X14.a (budget)
X15	Does the enriched gain survive frequency-only scoring under two splits?	Significant under both, inside budget	Same one scenario under both; 41 significant against 12 deployed	None
X17	Dual-duct AHU onboarded by configuration alone?	Config-only, gates clean, $\geq 60\%$ detected	45 of 55 at 3 false alarms of 96; one silence iteration	None; P6 failed (5 fouling, not ≤ 4)

Table 6: Every pre-registered experiment, its prediction, its outcome and what fired (part 1 of 2).

ID	Experiment	Prediction	Outcome	Falsifier
X18	Does the enriched-frequency gain replicate on the dual-duct unit?	A missed fouling scenario becomes significant	0 newly significant; 22 against 8 on already-detected	F-X18.b
X19 r1	Fan coil unit onboarded by configuration alone?	Gates clean, ≤ 10 false alarms, no conformance-only scenario	47 of 47 at 28 false alarms; 6 conformance-only	F-X19.b, F-X19.d
X19 A1	Does the amended onboarding hold?	Budget restored, no conformance-only detection	41 of 47 at 4 false alarms; 0 conformance-only	None
X20	Is model support the log's own heating-day count across four buildings?	Identity holds; $\rho < 0$ at $p < 0.05$	$\rho = -1$, $p = 1/24 = 0.042$; identity misses by 44 days on the fan coil unit	F-X20.a, F-X20.b
X21	Can a net's reading of a foreign year escape being that log's count?	Bound holds, tautology persists, ordering matches	Tautology fails on 6 of 12 pairs; foreign predictor $\rho = -0.6$	F-X21.a; P3 failed
X22	Does the conformance channel fire on injected order violations?	Reversal significant on the single-duct unit	Significant on 2 of 5 systems; single-duct hit rate 0 at zero false alarms	None; P1–P4 failed
X24	Does an outdoor-air-fraction residual add detections?	20 % damper leak detected, budget held	Fan coil 41 to 42 of 47 at 4 false alarms	None
X25 1–2	Are the conformance thresholds calibrated out of sample?	Rows out of sample, budget ≤ 10 , ≤ 3 verdict flips	Held at 0 flips; in-sample rows had understated false alarms by 3 and 4 days	None
X25 3	Does a max(fp,3)/n floor keep verdicts?	Counts within 2, ≤ 3 flips	2 flips; one gained, one lost	None
X25 4	Does a per-day residual null keep the detections?	Each count within 2	Series unit loses 3, 8 verdicts flip	F-X25.c
X26 (+A1)	Does the budget hold on a real building?	≤ 3 silence iterations, false alarms under 10%, fault case flagged ≥ 7 of 8 detected	6.1 % false-alarm days, every channel in budget; the 40 % undercharge not flagged	None; case predictions failed
X27	Does a fan-law virtual static recover static-pressure biases?		6 of 8; budget held	F-X27.b
X28	Does a water-side coil UA channel see the simulated fouling?	≥ 3 dual-duct and 2 fan coil fouling gained	0 gained; run 1 also broke the fan coil budget	F-X28.b; run 1 F-X28.a
X29	Is fouling visible when the refrigerant side is recorded?	≥ 3 of 5 condenser and evaporator severities, monotone	All 10 detected, monotone, 0 false alarms of 28	None
X30	Does any other process-mining instrument see the deployed misses?	No cell detects a miss within 3 false alarms	0 detections over 33 cells; worst cell 7 of 72	F-X30.b (two cells not evaluated)
X31 (+A1)	Field noise, change-of-value logging, schedule shifts?	Budget held; schedule arm within ± 1	Budgets held; dual-duct schedule arm 48 against 45	None; P3 and A1 failed
X33a–e	Does mapping every recorded column recover the misses?	No loss; budget $\leq$ clean +3	142 $\rightarrow$ 158 of 175, none lost, 21 $\rightarrow$ 26 of 480	None; three predictions failed upward, one downward
X34	Does the carrying rule name the seeded component?	Exact ≥ 60 ; exact or subsystem ≥ 85 ; wrong ≤ 10 %	Exact 69 %, subsystem 14 %, wrong 12.7 %, unnamed 7	F-X34.a; claim withdrawn
X35	Field conditions on the post-coverage configurations?	Budget $\leq$ clean +3; ≥ 2 edge recoveries lost under noise	Worst 7 of 96; at most one detection lost per cell; one edge recovery lost	None; P3 failed
X36	Do percentile bands keep the edge recoveries in budget?	False alarms rise ≤ 3 ; ≥ 2 new detections	Nothing gained or lost; +4 on both fan-powered units	F-X36.a
X37 (3 runs)	Do signed patterns and consensus diagnose family and component?	Family ≥ 80 %, unresolved ≤ 15 %; wrong < 10 %	Family 74.6 %, unresolved 27.8 %; wrong 10.1 %; dual-duct damper cases resolved	F-X37.a, F-X37.b
X38 (+A1)	Does the repaired Guideline 36 battery match STRATA under the same gate?	STRATA at least as many at lower false alarms	Silent-rule battery 12 of 14 @ 1 vs 13 @ 1; 8 of 30 and 9 of 29 @ 0 vs 26 and 25 @ 8 and 9	F-X38.a (G2, G3, S)